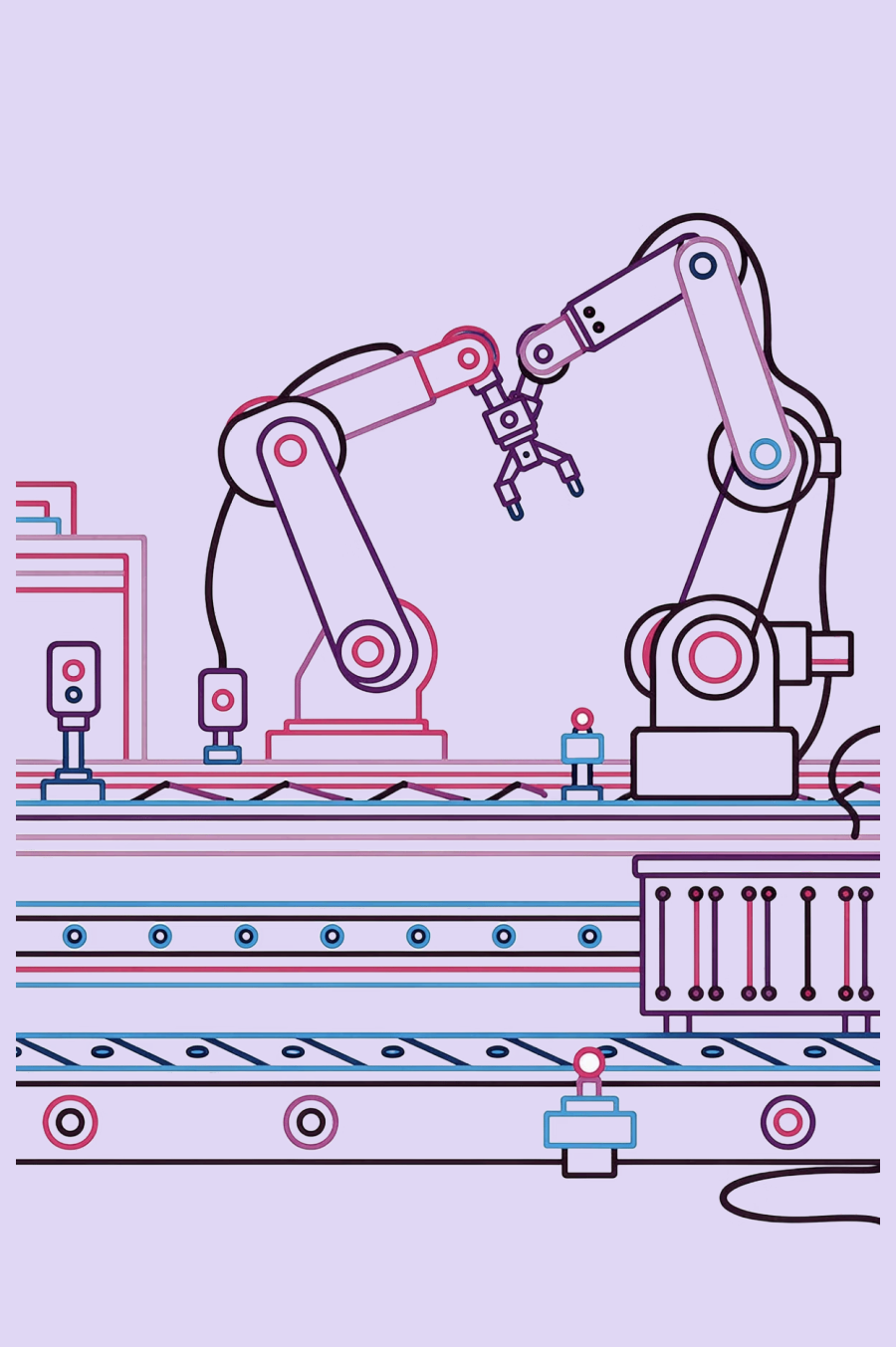




# Edge Computing in Industrial Communication

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# Introduction: Why Edge Computing?



With billions of devices online, sending all data to distant servers is no longer practical. Edge computing brings processing power **closer to the data source** — factory machines, pipeline sensors, or assembly line cameras. Only key results are sent to the cloud. Factories are time-critical: robotic arms and sensors must respond within **milliseconds**. A cloud reply can take over 500 ms — far too slow. Edge computing also provides **resilience**: if the cloud connection drops, the edge system operates independently [1, 2].

# Benefits & Application Areas

## Key Benefits

### Low Latency

Faster response for robotics, autonomous vehicles, emergency healthcare

### Reduced Bandwidth

Only filtered results sent to cloud, not raw IoT data streams

### Improved Privacy

Sensitive data stays local; only required data reaches the cloud

### Better Reliability

Operates even when cloud connection is poor or unavailable

## Application Areas



### IoT & Smart Cities

Smart homes, factories, traffic management



### Healthcare

Patient monitoring & emergency alerts



### Autonomous Vehicles

Local sensor & camera processing



### Industrial Automation

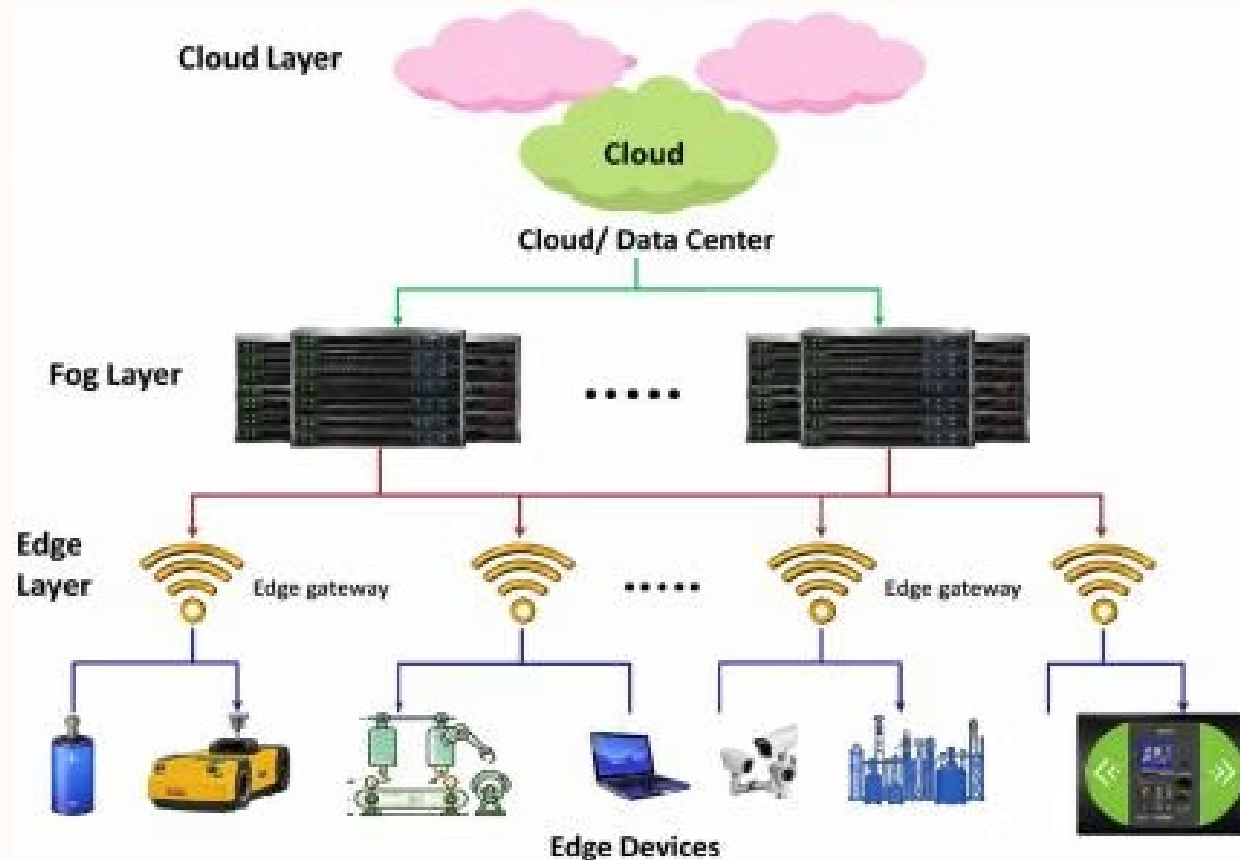
Fault detection & real-time machine control

# Basic Concept of Edge Computing



A computing node is placed **inside the industrial network**, near the machines, so data is processed immediately without involving the cloud. The edge node also acts as a **protocol translator** — for example, a MODBUS sensor and an OPC-UA robot cannot communicate directly, but an edge node bridges them. Only important results — alarms, status updates — are sent to SCADA or ERP systems. If a machine fails, the edge node detects the fault and sends a shutdown command within milliseconds [1, 2, 6].

# Architecture of Edge Computing



1

## Edge Layer

Gateways & local servers process raw machine data, translate protocols, run AI analytics

2

## Fog Layer

Buffers data between edge and cloud; prevents cloud overload

3

## Cloud Layer

Long-term storage, model retraining, ERP/MES reporting, global monitoring

Speed-sensitive tasks stay close to machines; complex analysis moves up the chain. **Cloudlets** and **Multi-Access Edge Computing (MEC)** provide cloud-like power without remote latency [2, 7].

# Edge vs. Cloud Computing

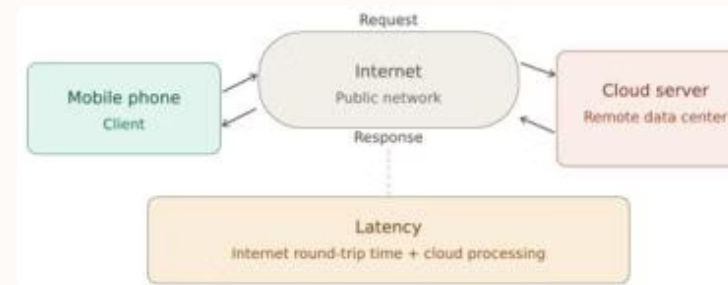
| Feature             | Cloud Computing             | Edge Computing          |
|---------------------|-----------------------------|-------------------------|
| Processing Location | Central cloud server        | Near the device or user |
| Latency             | Higher                      | Lower                   |
| Bandwidth Use       | More data sent to cloud     | Less data sent to cloud |
| Storage Capacity    | Very high                   | Limited                 |
| Best Use            | Big data storage & analysis | Real-time applications  |

The key difference is **where** data is processed. Cloud computing uses remote centralised servers; edge computing processes data near the user or device, reducing delay and network load [1, 2, 5].

# Latency Test: Setup & Method



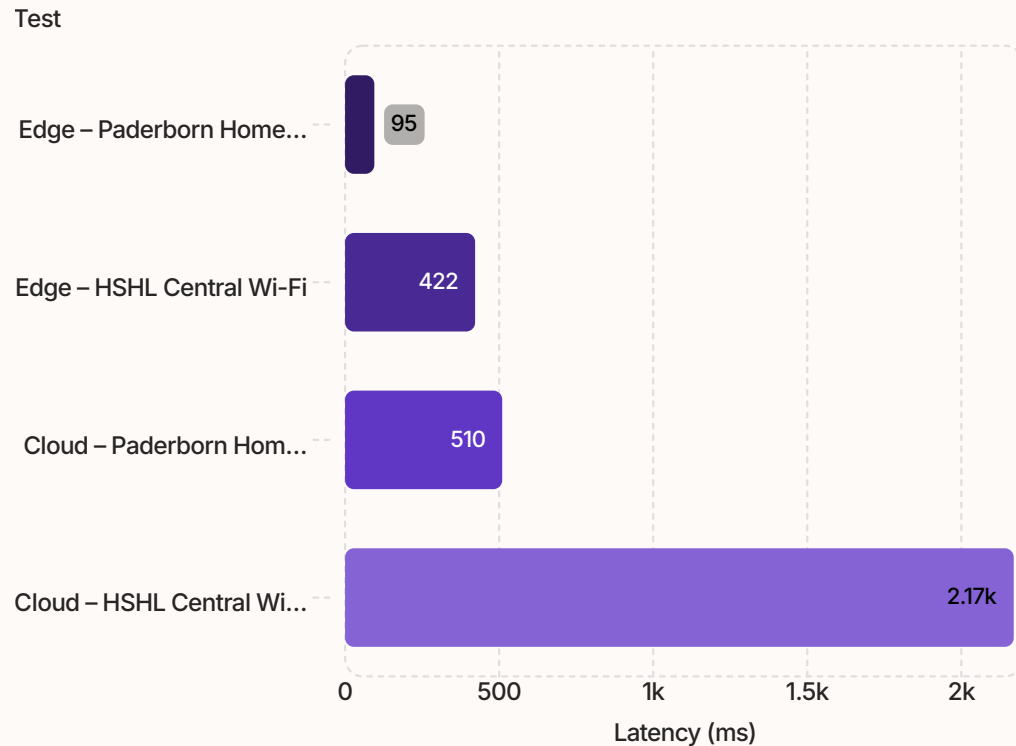
**Edge Test:** Mobile phone → Home Wi-Fi → Ubuntu 22.04 laptop (Flask server) → response. Local IP: 192.168.178.78:5000



**Cloud Test:** HTTP request sent to a remote server via the public internet. Measured using: `curl -w "%{time_total} seconds\n" -o /dev/null -s https://httpbin.org/get`

Latency was defined as the total round-trip time from request sent to response received. The latency formula used:  $C = \frac{X}{L} + \frac{D}{R}$ , where  $C$  = latency,  $X$  = distance,  $L$  = bandwidth,  $D$  = data rate,  $R$  = computing resources [8].

# Results & Comparative Analysis



## Key Findings

### 5.4× Faster

Edge (95 ms) vs. Cloud (510 ms) on stable home Wi-Fi

### 5.1× Faster

Edge (422 ms) vs. Cloud (2170 ms) on shared HSHL network

Shared networks slow both methods, but cloud is affected far more. A **dedicated local network** (private Ethernet or reserved wireless) is essential to achieve full edge performance advantages [9].



# Industrial Communication Mapping

| Experiment Component | Industrial Equivalent                           |
|----------------------|-------------------------------------------------|
| Mobile phone         | Sensor, PLC, robot, machine, or IIoT device     |
| Wi-Fi network        | Industrial Ethernet, PROFINET, OPC-UA, MQTT, 5G |
| Ubuntu laptop        | Edge gateway, industrial PC, or local server    |
| Flask application    | Local edge analytics or preprocessing software  |
| Latency value        | Communication response time in milliseconds     |

## PCB Assembly

Camera inspection rejects bad boards instantly

## CNC Machining

Edge detects tool wear before damage occurs

## Robotic Welding

Weld quality checked in real time; robot halted on defect

## Pharma Packaging

Underfilled or contaminated products removed on the spot

# Limitations & Challenges



## Security Issues

Edge devices are distributed across multiple sites, making them harder to protect than centralised cloud servers. A single compromised device can impact the entire system.



## Limited Resources

Edge devices have less computing power, memory, and storage than cloud servers. Applications must be carefully designed to fit within these constraints.



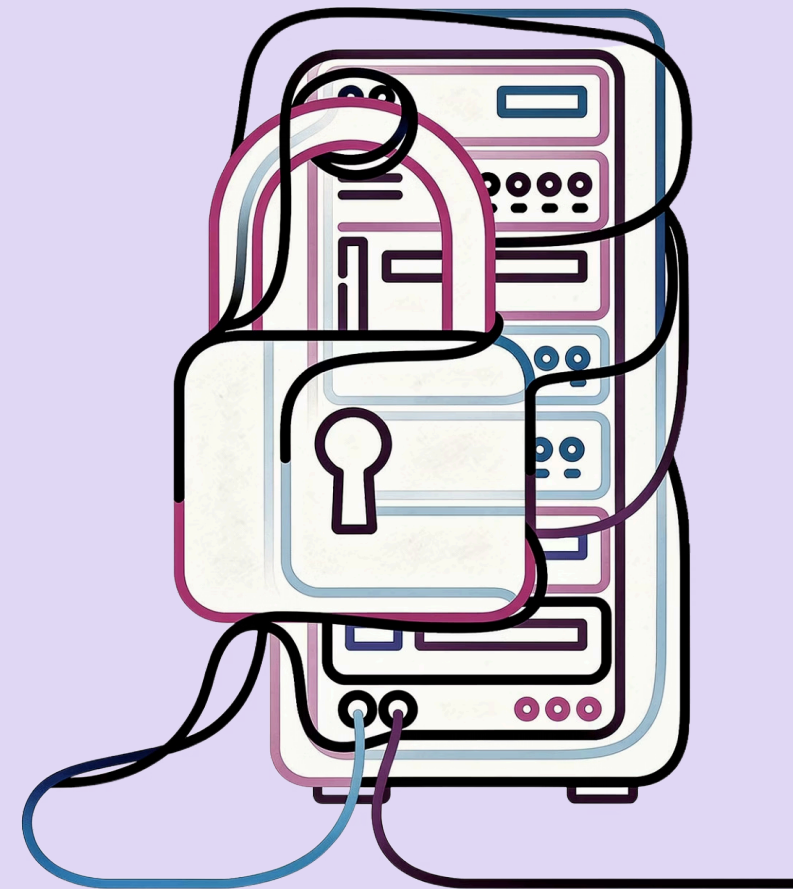
## Management Complexity

Managing many distributed edge devices is difficult. Software updates, monitoring, maintenance, and fault detection are all more complex at scale.



## Data Consistency

Data processed at different sites makes it difficult to maintain consistency between edge and cloud systems.



# Conclusion & References

## Conclusion

Edge computing is a key technology for modern distributed systems. It reduces latency, saves bandwidth, improves privacy, and supports real-time applications. Edge computing **complements** the cloud — the edge handles fast local processing while the cloud handles large-scale storage and analysis. Results confirm edge computing is **5.1–5.4× faster** than cloud in both stable and shared networks. A **hybrid architecture** is the practical solution for industry [9].

## Key References

- [1] Shi et al., "Edge Computing: Vision and Challenges," *IEEE IoT Journal*, 2016.
- [2] Satyanarayanan, "The Emergence of Edge Computing," *Computer*, 2017.
- [6] Fernandes et al., "Industrial IoT and Edge Computing," *IEEE Access*, 2022.
- [8] Cornelio & Cevallos-Torres, "Latency-Aware Edge Computing Framework," *Smart City Insights*, 2025.
- [9] Pallikonda et al., "Enhancing Performance in Autonomous Systems," *MITS*, 2025.
- [10] Bablu & Rashid, "Edge Computing for IoT-Driven Applications," *JACS*, 2025.

✓ Supervised by Prof. Dr.-Ing. Ali Hayek, Hochschule Hamm-Lippstadt